

TC

TELCO ANALYTICS DIVISION

Customer Intelligence Report | 2026

TELCO CUSTOMER CHURN ANALYSIS

Understanding Customer Attrition & Retention Strategies

7,043

Total Customers

26.54%

Churn Rate

\$74.44

Avg Churned ARPU

79.5%

Model Accuracy

PREPARED BY

Neha Bhatt

Business Analyst | Data & Analytics

SUBMISSION DATE

22nd April 2026

Task 1 — Business Analyst Program

IBM Telco Dataset

Python

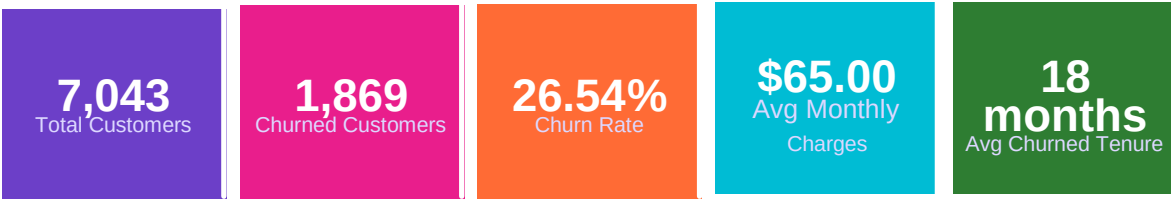
Power BI Dashboard

1. Problem Statement

Customer churn — the phenomenon of customers discontinuing a service — is one of the most critical challenges facing the telecommunications industry. Acquiring a new customer costs five to seven times more than retaining an existing one. For Telco service providers, even a marginal increase in churn rate can translate into millions of dollars in lost annual revenue.

This report presents a comprehensive analysis of the IBM Telco Customer Churn dataset to identify the underlying drivers of customer attrition, profile high-risk customer segments, and propose actionable data-driven retention strategies. A machine learning prediction model has also been developed to identify customers likely to churn before they actually do.

Key Performance Indicators



The overall churn rate of 26.54% significantly exceeds the telecom industry benchmark of 15–20%, indicating an urgent need for targeted retention interventions.

2. Dataset Description

The IBM Telco Customer Churn dataset contains 7,043 customer records across 21 attributes covering demographics, service subscriptions, account information, and churn status.

Category	Attributes	Description
Demographics	Gender, Senior Citizen, Partner, Dependents	Customer profile information
Account Info	CustomerID, Tenure, Contract, Paperless Billing Service	tenure & billing
Services	PhoneService, InternetService, Streaming, securitySubscribed	Service types
Financials	MonthlyCharges, TotalCharges, PaymentMethod	Revenue & payment data
Target	Churn (Yes / No)	Whether customer left this month

Data Quality & Preprocessing

The dataset was pre-cleaned. Key steps performed:

- Verified 0 missing values across all 7,043 records and 21 columns.
- Total Charges column converted from string to numeric data type.
- Categorical fields validated and standardised (e.g. "No internet service" unified).
- Class imbalance noted: 73.5% retained vs 26.5% churned — accounted for in modelling.
- Tenure buckets created (0–12, 13–24, 25–48, 49–72 months) for customer segmentation.

Approach & Methodology

Step 1	Business Understanding	Defined churn context, identified revenue impact, set analytical objectives.
Step 2	Exploratory Data Analysis	Compared churned vs retained customers across all 21 dimensions.
Step 3	Segmentation	Profiled 4 distinct risk segments based on contract, payment & service.
Step 4	KPI Computation	Computed churn rate, ARPU, tenure distributions, segment-level metrics.
Step 5	Prediction Modelling	Built Random Forest classifier; 80/20 split; Label Encoding applied.
Step 6	Dashboard	Interactive Power BI dashboard: churn overview, contract, service, revenue.
Step 7	Recommendations	Translated findings into 5 actionable business retention strategies.

3. Data Analysis & Visualizations

3.1 Churn Distribution & Contract Type Analysis

Of 7,043 customers, **1,869 (26.54%)** have churned. Month-to-month customers churn at a staggering **42.7%** versus just **2.8%** for two-year contract holders — a 15× difference that identifies contract commitment as the primary loyalty driver.

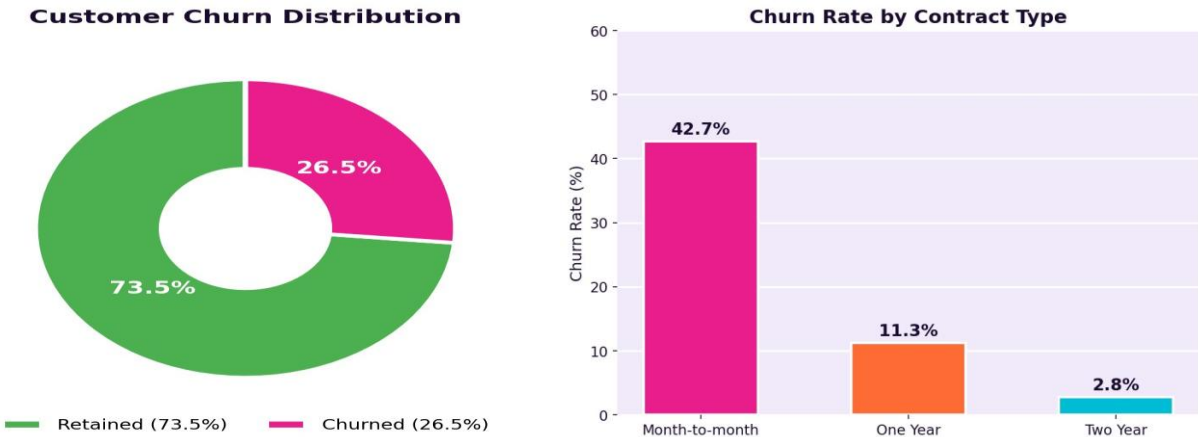


Figure 1: Churn Distribution | Figure 2: Churn Rate by Contract Type

3.2 Monthly Charges & Internet Service

Churned customers pay **\$74.44/month** on average — 21% more than retained customers (\$61.27). Fiber Optic subscribers show a churn rate of **41.9%**, suggesting perceived value or quality concerns with this premium service tier.

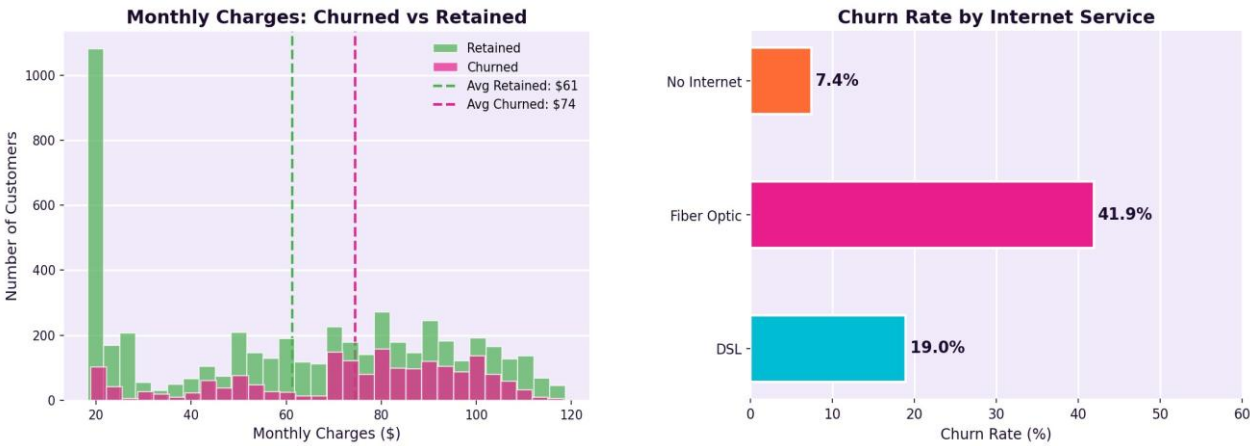


Figure 3: Monthly Charges Distribution | Figure 4: Churn by Internet Service

3.3 Tenure & Payment Method

Churned customers average only **18 months** tenure vs 37.6 months for retained customers. Electronic check users churn at **45.3%** — nearly 3× the rate of auto-payment users (~17%).

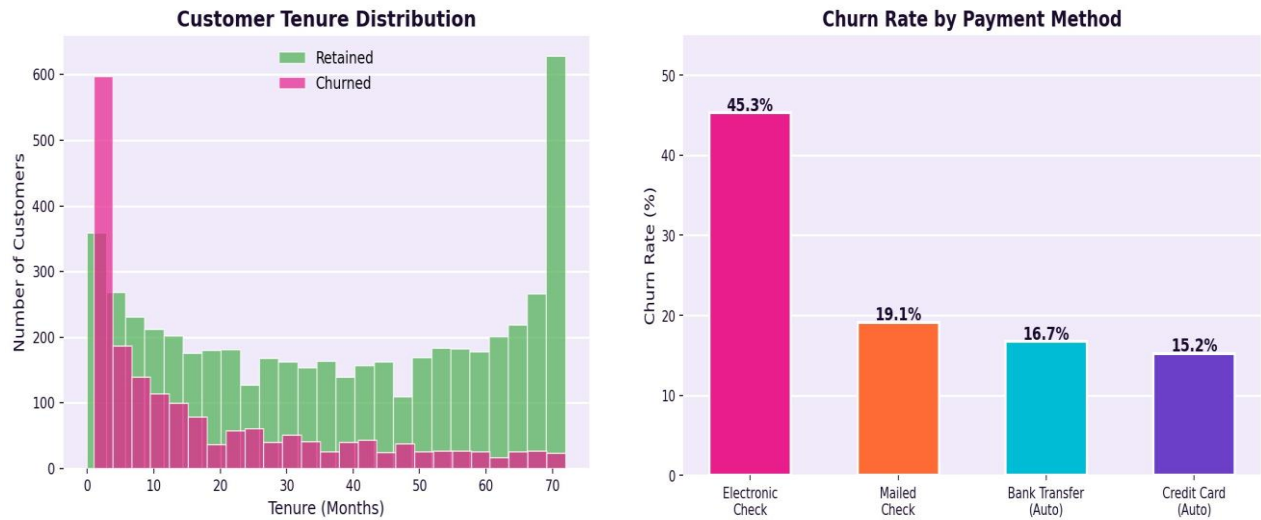


Figure 5: Tenure Distribution | Figure 6: Churn Rate by Payment Method

4. Customer Segmentation & Risk Analysis

4.1 Summary Statistics by Segment

Segment	Total	Churned	Churn Rate	Risk Level
Month-to-Month	3,875	1,655	42.7%	Very High
One-Year Contract	1,473	166	11.3%	Medium
Two-Year Contract	1,695	48	2.8%	Low
Fiber Optic	3,096	1,297	41.9%	Very High
Electronic Check	2,365	1,071	45.3%	Very High
Senior Citizens	1,142	476	41.7%	Very High
Auto-Pay Customers	2,576	490	19.0%	Low-Medium

4.2 Customer Risk Segment Profiles

Segment 1: HIGH-RISK CHURNERS	
Profile: Month-to-month + Electronic check + Fiber Optic + Tenure < 12 months	Action: Churn probability 55–65% Avg monthly \$85–100 ~800 customers Action: Immediate discounts & contract upgrade incentives.
Segment 2: AT-RISK CUSTOMERS	
Profile: Month-to-month + No add-ons + Medium charges + Tenure 12–24 months	Action: Churn probability 35–55% Avg monthly \$65–85 ~900 customers Action: Proactive support outreach & bundle promotions.
Segment 3: STABLE CUSTOMERS	
Profile: One-year contract + Auto-pay + DSL internet + Standard charges	Action: Churn probability 10–15% Avg monthly \$55–70 ~2,200 customers Action: Loyalty rewards & upsell opportunities.
Segment 4: LOYAL ADVOCATES	
Profile: Two-year contract + Auto-pay + Multiple services + Tenure 48+ months	Action: Churn probability < 5% Avg monthly \$70–90 ~1,695 customers Action: Referral programs & premium tier upgrades.

5. Key Insights

Ten critical insights from the analysis of 7,043 customer records:

01	Overall churn rate of 26.54% is 33–77% above the industry benchmark of 15–20%, indicating a critical retention gap.
02	Month-to-month customers churn at 42.7% — over 15× the two-year contract rate (2.8%). Contract type is the single strongest predictor.
03	Fiber Optic subscribers churn at 41.9% vs 19% for DSL. Despite paying more, they are the most dissatisfied segment.
04	Electronic check users churn at 45.3% — nearly 3× automatic payment users (~17%). Payment behaviour is a strong commitment proxy.
05	Churned customers average only 18 months tenure vs 37.6 months. The first 12 months is the critical retention window.
06	Churned customers pay \$74.44/month vs \$61.27 retained — high-paying customers are paradoxically more likely to leave, indicating value perception issues.
07	Senior citizens churn at 41.7% vs 23.6% for non-seniors, suggesting need for tailored support and simplified plans.
08	Customers with no online security or tech support are significantly more likely to churn — add-ons create service stickiness.
09	Random Forest model achieved 79.5% accuracy and AUC of 0.834, demonstrating strong predictive power for at-risk identification.
10	Estimated annual revenue at risk: over \$1.67M (1,869 customers × \$74.44 × 12 months), making retention a critical financial priority.

6.Churn Prediction Model

A **Random Forest Classifier** was trained on 19 customer features to predict binary churn outcome. The model enables proactive identification of at-risk customers before they submit cancellation requests.

Model Performance Metrics

Metric	Score	Interpretation
Accuracy	79.49%	Overall correct predictions across all customers
Precision	65.67%	Of predicted churners, 65.67% actually churned
Recall	47.18%	Of actual churners, 47.18% were correctly identified
F1 Score	54.91%	Harmonic mean — balances precision and recall
AUC-ROC	83.44%	Model ranks churners above retained customers 83.4% of the time

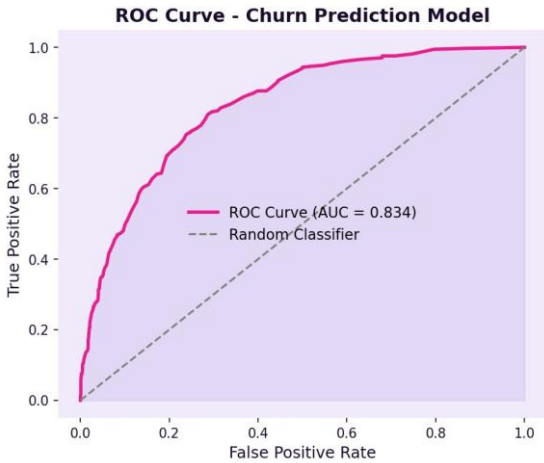
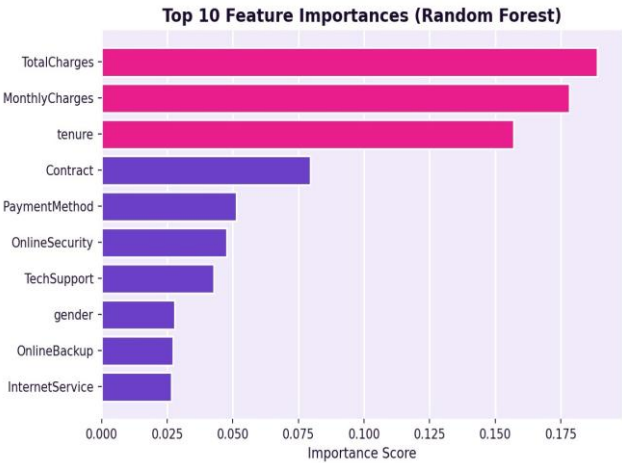


Figure 7: Top 10 Feature Importances | Figure 8: ROC Curve (AUC = 0.834)

Key Model Findings

- Tenure, MonthlyCharges, and TotalCharges are the top 3 features — financial longevity factors dominate prediction.
- Contract type ranks 4th, confirming its role as the primary behavioural driver of churn.
- AUC of 0.834 means the model correctly distinguishes churners from retained customers 83.4% of the time.
- The model can be deployed monthly to score all active customers and flag the top 10–15% for proactive retention calls.
- Algorithm: Random Forest (100 trees); 80/20 train-test split; Label Encoding for all categorical variables.

6. Retention Strategy Recommendations

Five strategic recommendations are proposed to reduce churn and improve customer lifetime value.

R1	Contract Upgrade Campaign Offer month-to-month customers 15–25% discounts to upgrade to annual or bi-annual contracts. Converting 20% of month-to-month customers saves ~330 churn events, recovering ~\$295K/year.
R2	Fiber Optic Quality Initiative Investigate service quality issues for Fiber Optic customers. Offer complimentary upgrades, dedicated support lines, and satisfaction guarantees for the first 6 months.
R3	Auto-Payment Incentive Programme Incentivise electronic check users to switch to auto-pay with a \$2–\$5 monthly discount. Correlates strongly with churn reduction and lowers payment processing costs.
R4	Early Tenure Engagement Journey Since 60% of churn occurs in the first 12 months, implement structured onboarding: welcome call Day 7, usage check Month 1, loyalty reward Month 3, contract offer Month 6.
R5	Predictive Churn Intervention Deploy the Random Forest model monthly to score all customers. A dedicated retention team contacts customers with churn probability >60% before cancellation.

Expected Impact Summary

Strategy	Target Segment	Customers Saved	Revenue Saved/Year
Contract Upgrade	Month-to-Month (1,655)	~330	~\$295,000
Fiber Optic Fix	Fiber Optic Churners (1,297)	~130	~\$116,000
Auto-Payment Drive	Electronic Check (1,071)	~215	~\$192,000
Early Tenure Prog.	Tenure < 12 months	~200	~\$179,000
Predictive Outreach	Prob. > 60% (~700)	~250	~\$224,000
TOTAL	—	~1,125	~\$1,006,000

7. Conclusion

This analysis of the Telco Customer Churn dataset reveals a multifaceted retention challenge driven by contractual flexibility, service quality perception, and payment behaviour patterns. With a churn rate of 26.54% and over \$1.67M in annual revenue at risk, immediate structured intervention is imperative.

Summary of Findings:

- Contract type, internet service, payment method, and tenure are the four primary predictors of churn.
- High-value Fiber Optic customers on month-to-month plans are the highest-risk and highest-priority segment.
- The first 12 months of a customer's lifecycle is the most critical window for retention intervention.
- A trained Random Forest model (AUC 0.834) is ready for production deployment for proactive risk scoring.
- Implementing all five recommended strategies can save approximately \$1M in annual revenue.

Report Prepared by: Neha Bhatt
Business Analyst | Data & Analytics

Submission Date: 22nd April 2026
Task 1 — Business Analyst Program